Development of Ferritic Stainless Steels for Use in Desalination Plants*

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Abstract

This paper describes a research program aimed at optimizing the composition, heat treatment, and structure of ferritic stainless steels for use as condenser tube material in desalination plants. The results of corrosion tests carried out in a desalination plant are reported and discussed in relation to laboratory tests performed on the same alloy compositions. The corrosion measurements are used to outline general methods for maximizing the resistance of ferritic stainless steels to pitting and crevice crack attack in hot saline environments.

The earlier work carried out on this program[(1)] has been published previously.[1,2] It is the purpose of this paper to discuss the more recent results and to review certain aspects of the earlier work in terms of the potential application of ferritic stainless steel condenser tubes in multi-stage flash evaporator desalination plants.

Materials

Alloy Selection

The compositions selected for laboratory tests and for desalination plant exposure tests are listed in Tables 1-3. The selection of the compositions in Table 1 for desalination plant testing was based on a previous rapid screening of many ferritic stainless steels, which showed that the chosen alloys would represent a range of pitting resistance defined by pitting potential values between 100 and 500 mV vs SCE when measured in deaerated synthetic seawater at 90 C (194 F) and pH = 7.2 ± 0.2.

In addition to the compositions in Table 1 which were prepared by vacuum induction melting of 50 lb heats, additional compositions (50 gm levitation melts) were also prepared with the prime objective of relating composition, pitting potential, and resistance to crevice attack (Tables 2 and 3). The nominal compositions given in Table 3 were based on a series of Fe-Cr-Mo alloys in which partial substitutions for Cr of (Si + Ni) or (Al + Ni) and of (Nb + Mn) for Mo, had been made.

Melting Procedures and Sample Preparation

The 50 gm heats were prepared by levitation melting as described previously,[1] to produce ingots in the form of flat slabs (¼ x 1 x 1½ inches). All surfaces of the cast slabs were ground to remove imperfections and provide good rolling surfaces before annealing 16 hours at 1200 C (2192 F) in argon atmosphere. The slabs were hot rolled at 1090 C

TABLE 1 — Alloy Compositions for Initial Tests in Freeport Desalination Plant

	Composition Number	Alloying Additions (Listed as Wt %)					Interstitial Content (ppm)				
		Fe	Cr	Mo	Si	Ni	C	O	S	N	P
W5	Aim	72.73	21.48	3.00	1.36	1.43	<50	<300	<100	<100	<50
	Actual-Bottom	Bal.	21.50	2.90	1.26	1.47	39	220	80	66	10
	Actual-Top	Bal.	21.40	2.50	1.26	1.48	36	230	81	67	11
W6	Aim	73.66	21.51	2.03	1.36	1.44	<50	<300	<100	<100	<50
	Actual-Bottom	Bal.	21.60	2.00	1.29	1.46	25	250	79	60	12
	Actual-Top	Bal.	21.60	1.56	1.28	1.47	41	960	81	60	13
W7	Aim	71.70	21.50	4.00	1.36	1.44	<50	<300	<100	<100	<50
	Actual-Bottom	Bal.	20.60	4.20	1.32	1.50	34	230	77	54	12
	Actual-Top	Bal.	20.60	4.27	1.32	1.51	35	300	90	59	10
W8	Aim	69.20	23.00	5.00	1.36	1.44	<50	<300	<100	<100	<50
	Actual-Bottom	Bal.	23.00	5.12	1.30	1.49	42	290	81	62	15
	Actual-Top	Bal.	23.10	5.19	1.26	1.47	46	270	85	57	20

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TABLE 2 – Alloys Selected for Study In Fe-Cr-Mo Ternary System

Identity	Composition (Wt %)
F389B	Fe-12 Cr
F390B	Fe-15 Cr
F294B	Fe-20 Cr
F314B	Fe-25 Cr
F190B	Fe-30 Cr
F191B	Fe-35 Cr
F307B	Fe-40 Cr
F308B	Fe-14 Cr-1 Mo
F309B	Fe-14 Cr-3 Mo
F310B	Fe-14 Cr-5 Mo
F311B	Fe-17 Cr-1 Mo
F312B	Fe-17 Cr-3 Mo
F313B	Fe-17 Cr-5 Mo
F295B	Fe-20 Cr-1 Mo
F296B	Fe-20 Cr-3 Mo
F297B	Fe-20 Cr-5 Mo
F291B	Fe-25 Cr-1 Mo
F292B	Fe-25 Cr-3 Mo
F293B	Fe-25 Cr-5 Mo
F223B	Fe-30 Cr-1 Mo
F224B	Fe-30 Cr-3 Mo
F230B	Fe-23.2 Cr-1.75 Mo
F231B	Fe-23.4 Cr-3.5 Mo
F232B	Fe-23.2 Cr-5.07 Mo

(1994 F) to 1/16 inch thickness. Oxide scale was subsequently removed by sand blasting before a final annealing treatment. Discs were spark cut from the resulting strips for both polarization measurements and for crevice corrosion studies.

The 50 pound ingots, in slab form, were cast directly into a heated graphite ingot mold. The mold was a split design having internal measurements of 2 x 6 x 16½ inches, including a 2 inch tapered hot top.

Following preparation, each alloy was hot rolled to 0.100 inch sheet and water quenched. Hot rolling temperatures were restricted to a maximum of 982 C (1800 F) and a minimum of 816 C (1500 F). The hot rolled sheet was electropolished in an aqueous 15 wt% NaCl/30 wt% H_2SO_4 electrolyte [using a current density of 1200 amps/ft^2 for 5 minutes at 60 C (140 F)] to remove surface scale and sheared into 1 inch wide strips running parallel to the hot rolling direction. The 1 inch wide hot rolled strips were subsequently cold rolled from 0.100 inch to a final thickness of 0.0625 inch. The finished strips were sheared into 6½ inch lengths and machined to a final size of 6 x 3/4 inch to provide test coupons for the OSW Freeport Desalination Plant Test Facility.

To allow studies of the effects of metallurgical structure, test coupons were prepared for each composition in Table 1 in four different conditions. The specific treatments are listed in Table 4.

Corrosion Measurements

Pitting Resistance

Experimental Procedures. In the present work, electrochemical test methods were used to evaluate the relative pitting resistance of the ferritic steels. All measurements were made in deaerated synthetic seawater at 90 C.

TABLE 3 – Alloys Selected for Laboratory Studies (50 gm Ingots)

Identity	Fe	Cr	Mo	Si (Wt %)	Ni	Al	Nb	Mn	Basic Composition (Wt %)	Elements Replaced (At %)
F460	73.26	24.32	1.02	0.68	0.72	–	–	–	Fe-26Cr-1Mo	2Cr
F461	73.63	22.53	1.02	1.37	1.45	–	–	–	Fe-26Cr-1Mo	4Cr
F462	74.00	20.73	1.03	2.07	2.17	–	–	–	Fe-26Cr-1Mo	6Cr
F463	73.49	22.56	0.71	1.37	1.44	–	0.34	0.09	Fe-26Cr-1Mo	4Cr+0.2Mo
F464	73.37	22.52	0.70	–	1.85	1.13	0.34	0.09	Fe-26Cr-1Mo	4Cr+0.2Mo
F465	73.30	23.29	2.02	0.68	0.71	–	–	–	Fe-25Cr-2Mo	2Cr
F466	73.66	21.51	2.03	1.36	1.44	–	–	–	Fe-25Cr-2Mo	4Cr
F467	74.03	19.71	2.04	2.06	2.16	–	–	–	Fe-25Cr-2Mo	6Cr
F468	73.48	21.53	1.75	1.37	1.44	–	0.34	0.09	Fe-25Cr-2Mo	4Cr+0.2Mo
F469	73.37	21.49	1.75	–	1.84	1.12	0.34	0.09	Fe-25Cr-2Mo	4Cr+0.2Mo
F470	72.73	21.48	3.00	1.36	1.43	–	–	–	Fe-25Cr-3Mo	4Cr
F471	73.10	19.69	3.02	2.05	2.14	–	–	–	Fe-25Cr-3Mo	6Cr
F472	72.65	21.51	2.62	1.36	1.43	–	0.34	0.09	Fe-25Cr-3Mo	4Cr+0.2Mo
F473	73.02	19.70	2.64	2.06	2.15	–	0.34	0.09	Fe-25Cr-3Mo	6Cr+0.2Mo
F474	71.93	19.74	3.50	2.05	2.14	–	0.51	0.13	Fe-25Cr-4Mo	6Cr+0.3Mo
F475	70.97	19.73	4.70	2.04	2.13	–	0.34	0.09	Fe-25Cr-5Mo	6Cr+0.2Mo

TABLE 4 – Pitting Potential Measurements on Alloy Compositions Used in Desalination Plant Exposure Tests

Identity	E_c^{scan} (mV vs SCE)	E_c^{scr} (mV vs SCE)	Specimen Condition,[(1)] Heat Treatment
WE02 (E-Brite 261)	242	155	As rolled
W502	662	310	As rolled
W602	90-388	215	As rolled
W702	830	425	As rolled
W802	706	565	As rolled
W582	160-642	300	800 F x 1 hr, chamber cool
W682	472	205	800 F x 1 hr, chamber cool
W782	285-460	345	800 F x 1 hr, chamber cool
W882	550	530	800 F x 1 hr, chamber cool
WE62 (E-Brite 261)	288	140	1450 F x 1 hr, chamber cool
W549	420	350	2000 F x 1 hr, water quench
W649	288	220	2000 F x 1 hr, water quench
W749	410	405	2000 F x 1 hr, water quench
W849	580	595	2000 F x 1 hr, water quench

[(1)]All specimens pickled 5 minutes in 10% HNO_3 + 0.5% HF, 85 C prior to testing in deaerated, synthetic seawater, 90 C, pH = 7.2 ± 0.2.

Before using the seawater in the polarization experiments, the solution was treated with sulfuric acid to neutralize bicarbonate alkalinity and subsequently boiled to remove free carbon dioxide and dissolved oxygen. The pH was adjusted to 7.2 ± 0.2 at 90 C. The polarization cells and associated equipment which were used for the electrochemical measurements have been described previously.[1-3]

Three different experimental procedures have been used to determine the critical pitting potentials of the ferritic steels. The first method involved a conventional potentiodynamic anodic polarization of the specimen at 10 mV/min until a sudden increase in specimen current indicated the pitting potential, E_c^{scan}. As described in our previous work,[3] the value of E_c^{scan} is a function of the experimental procedure, surface finish, and polarization rate. Consequently, the E_c^{scan} value was used only as a reference point for subsequent tests.

The second experimental procedure involved determination of the *scratch potential*, E_c^{scr}. The technique for making E_c^{scr} measurements has been described previously.[3] The E_c^{scr} value is essentially independent of both surface finish and polarization rate and is therefore an excellent means of distinguishing the intrinsic pitting resistances of a series of alloys.

In order to substantiate the E_c^{scr} value, current-time curves were also determined for some alloys at potentials between E_c^{scr} and E_c^{scan}. In these experiments, pitting was observed to occur at potentials below E_c^{scan} but rarely below E_c^{scr}. Comparison of the pitting resistances of the alloys was therefore based solely on E_c^{scr} measurements.

Experimental Results. Figure 1 illustrates the pitting resistances, as defined by E_c^{scr}, of a wide range of Fe-Cr-Mo ternary alloys. Included in Figure 1 are the corresponding E_c^{scr} values for several commercial stainless steels and nickel base alloys. The broken lines, interpolated from the experimentally determined solid curves, have been included to indicate the expected behavior of ferritic alloys with the same (Cr + Mo) content as the commercial alloys. It is particularly interesting, therefore, to note the coincidence of the pitting resistance of Ni containing commercial alloys with the broken curves in Figure 1, which indicate that the pitting resistance of such alloys is dictated almost entirely by the (Cr + Mo) content. Inconel 625, for example, with a composition of Fe-22.8Cr-9.1Mo-60.5Ni-3.6(Nb + Ta) exhibits the same E_c^{scr} value as would be expected for a ternary ferritic alloy of Fe-22.8Cr-9.1 Mo.

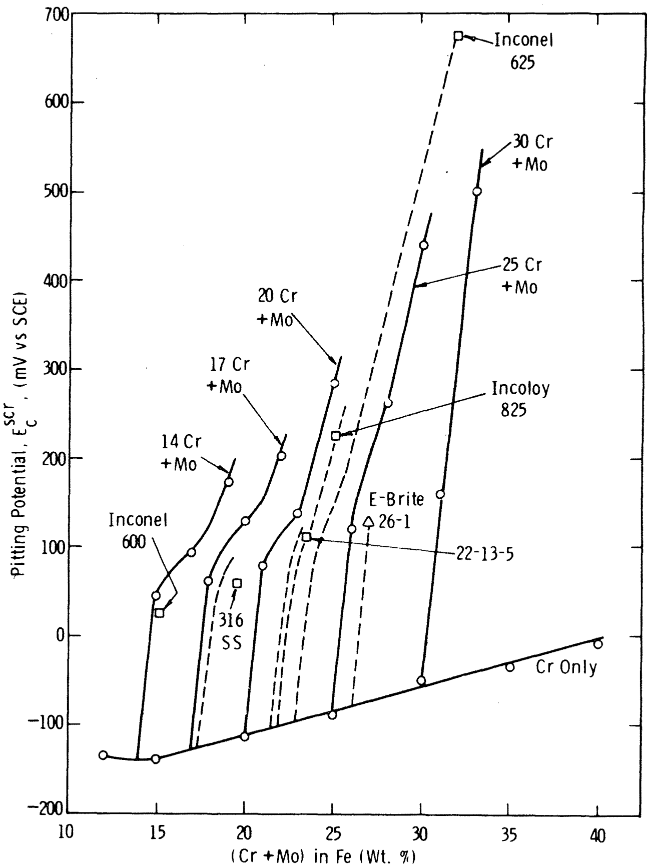


FIGURE 1 – Comparison of the critical pitting potentials of Fe-Cr-Mo alloys with several commercial alloys in deaerated synthetic seawater at 90 C, pH = 7.2 ± 0.2.

The results of additional anodic polarization measurements carried out in deaerated, synthetic seawater at 90 C are presented in Table 5 and illustrated in Figure 2. Fourteen ferritic steels were studied and a specimen of E-Brite 26-1 also included as reference material.

Figure 2 illustrates the effects of five different alloy additions to various Fe-Cr-Mo base compositions. The lower

TABLE 5 – Results of Pitting Potential Measurements on Ferritic Stainless Steels

Identity	Run No.	E_c^{scan} (mV vs SCE)	E_c^{scr} (mV vs SCE)	$E_c^{(l-t)}$ (mV vs SCE)
E-Brite	1131	136	128	<135
F460B	1108	174	145	–
F461B	1120	136	108	–
F462B	1117	184	178	–
F463B	1121	194	186	–
F464B	1134	142	98	<105
F464B	1123	125	98	<100
F465B	1129	192	170	<175
F466B	1135	262	232	<240
F467B	1138	226	215	<220
F468B	1112	194	192	–
F469B	1113	284	165	<280
F470B	1136	306	272	–
F471B	1139	230	222	<225
F472B	1142	225	174	–
F473B	1140	240	235	<238
F474B	1141	342	268	<330
F475B	1143	310	298	–
F461A	1204	152	142	–
F462A	1152	190	165	<170
F463A	1160	286	174	<190
F464A	1150	140	95	<130
F466A	1148	238	230	–
F467A	1205	218	185	–
F469A	1145	276	270	<275
F470A	1158	256	248	<250
F473A	1207	182	165	–
F474A	1154	220	205	<206
F475A	1190	238	152	<160

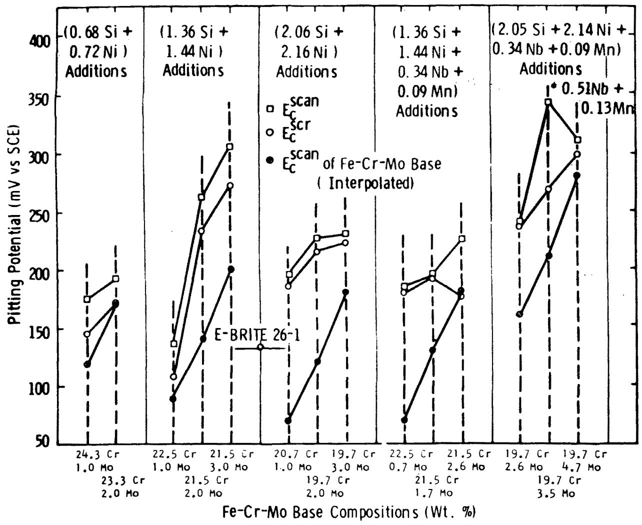


FIGURE 2 — Critical pitting potentials of levitation melted alloys in deaerated synthetic seawater of 90 C, pH = 7.2 ± 0.2.

curves in each alloy group (filled circles) indicate the E_c^{scan} values of the ternary base compositions. The improvements in pitting resistance, produced by the additions, are indicated by the top curves (open circles and squares). It is evident that (Si + Ni) alloy additions are capable of substantially improving the pitting resistance of certain Fe-Cr-Mo alloys. All data presented in Figure 2 were obtained using B-series alloys which had received a final heat treatment of 1 hour at 1049 C (1920 F) followed by water quenching.

Using similar experimental techniques, the relative pitting resistance of the five alloy compositions prepared for exposure at Freeport have been determined as a function of heat treatment. The compositions are given in Table 1, and the experimental data listed in Table 4.

As shown in Figure 3, the critical pitting potential measurements indicated a rapid improvement in pitting resistance with increasing molybdenum content, i.e., E-Brite 261 (1%Mo)-W6 (2%Mo)-W5 (3%Mo)-W7 (4%Mo)-W8 (5%Mo). This group of alloys, which exhibits such a broad range of pitting potentials, therefore provides an excellent opportunity to correlate E_c^{scr} measurements with the behavior of the alloys under actual desalination conditions. By comparing the E_c^{scr} data in Figure 3 with the results of desalination plant exposures of the same alloys prepared in an identical manner (Figures 4, 5), a qualitative correlation is, in fact, apparent.

Crevice Attack Resistance. In addition to the relative pitting resistance of the alloys, the possibility of crevice attack is also an important problem in desalination plant operation. This mode of attack can occur at tube connections or under deposits of sand, shells, or other fouling materials. It follows, therefore, that in choosing materials for equipment with unavoidable recesses and crevices, it is essential to have a knowledge of their relative susceptibilities to crevice attack.

Two laboratory methods have been used to evaluate relative resistances to crevice attack. These involved exposure of test specimens either to an oxygenated, 10% ferric

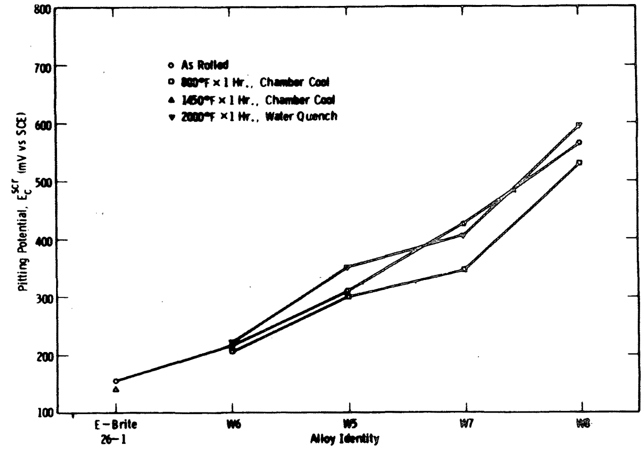


FIGURE 3 — Critical pitting potentials of specimens prepared identically to those used in the desalination plant tests. Measurements made in deaerated synthetic seawater at 90 C, pH = 7.2 ± 0.2.

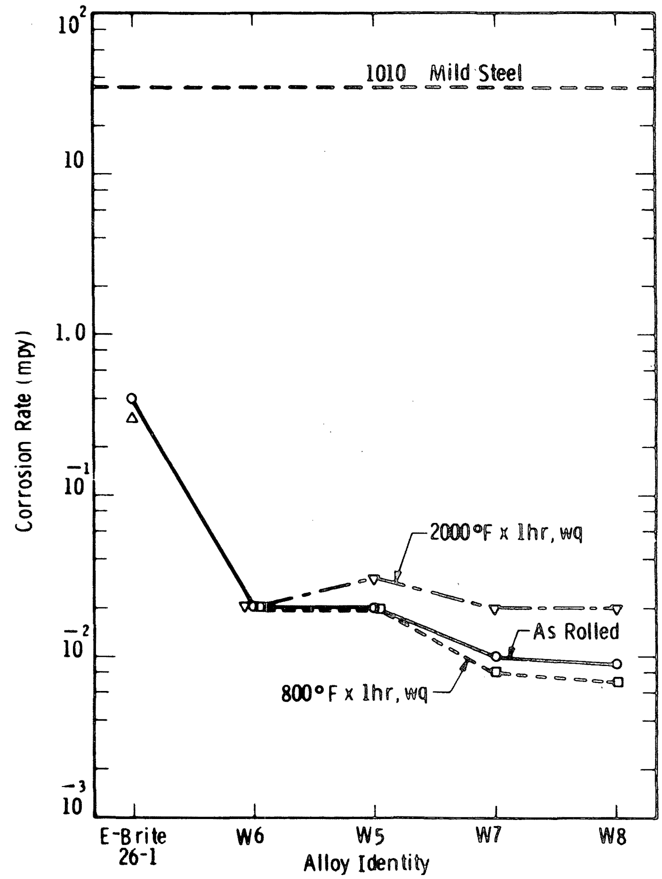


FIGURE 4 — Corrosion rates after Run 1 in desalination plant.

chloride solution at ambient temperature or to synthetic seawater at 250 C (482 F) with various levels of dissolved oxygen. In both cases, crevice conditions were deliberately introduced at the surface of all specimens in the manner illustrated in Figure 6.

Some results of the crevice attack studies, carried out in oxygenated, 10% ferric chloride solution, are present in Figures 7 and 8. In Figure 7 the specimens are arranged in order of decreasing pitting potential. Although there is not

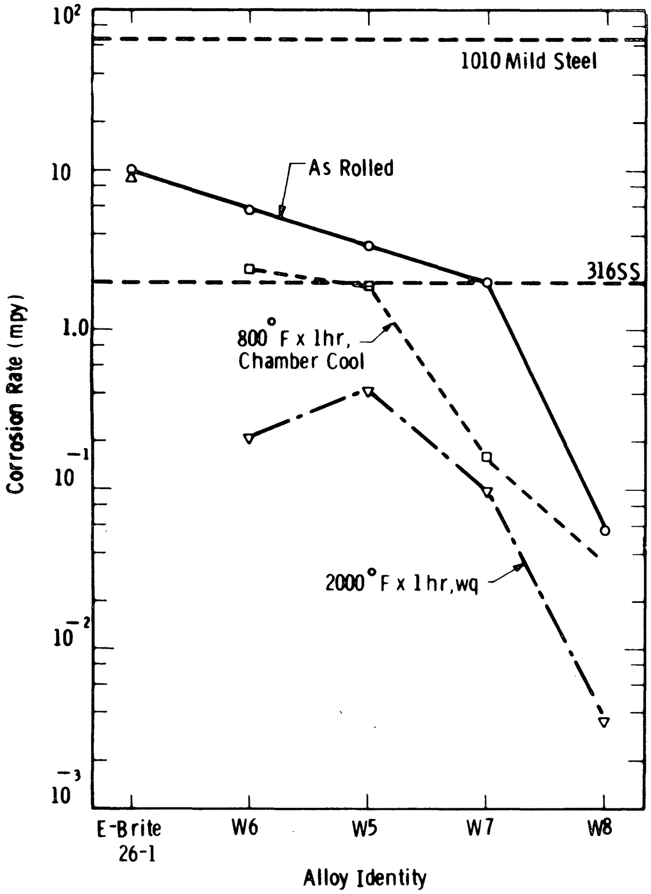


FIGURE 5 — Corrosion rates after Run 5 in desalination plant.

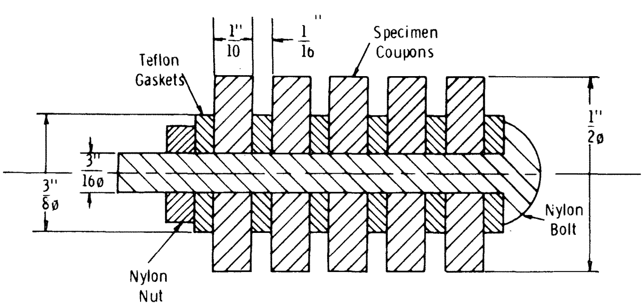


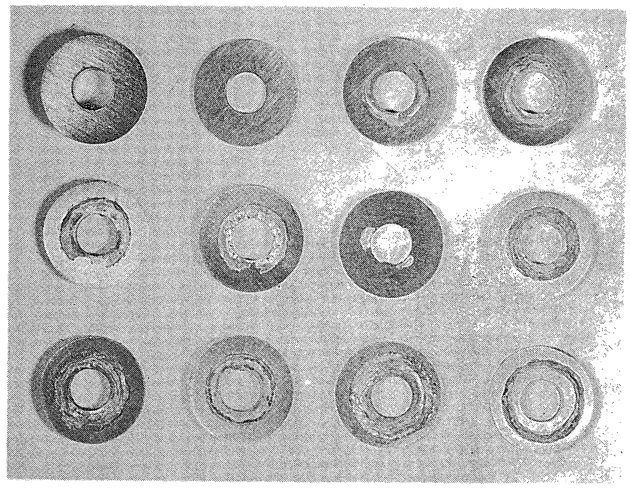
FIGURE 6 — Experimental arrangement of specimens in crevice attack studies.

a one to one correlation between pitting potential and degree of crevice attack in the ferric chloride solution, there is a clear tendency for the extent of crevice attack to be reduced as the pitting potential is increased.

Figure 8 illustrates the effect of heat treatment on the resistance to crevice attack of several of the ferritic stainless steels. It is noteworthy that those alloys which exhibit the most marked influence of heat treatment on crevice attack resistance are also the alloys which exhibit a correspondingly marked change in critical pitting potential.

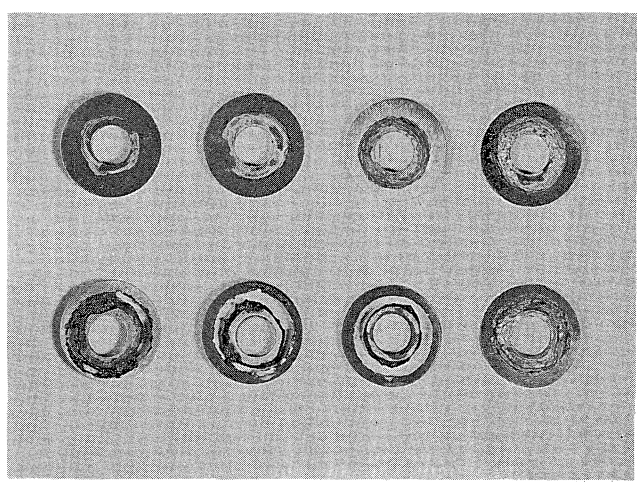
The results of ferric chloride tests on the ternary Fe-Cr-Mo alloys are presented in Figures 9 and 10. The alloys F308B, F309B, F389B, F390B, and F294B are not included in Figures 9 and 10 because all of these alloys completely disintegrated after the 6 day exposure.

The corresponding results of a severe 2 week direct



Ti	F232B E_c=410	F475B E_c=298	F470B E_c=272
F474B E_c=268	F469B E_c=265	F231B E_c=240	F473B E_c=235
F466B E_c=232	F471B E_c=222	F467B E_c=215	F468B E_c=192

FIGURE 7 — Crevice corrosion after 6 day exposure to oxygenated 10% ferric chloride at 21 ± 1 C. Specimen size: ½ inch diameter.



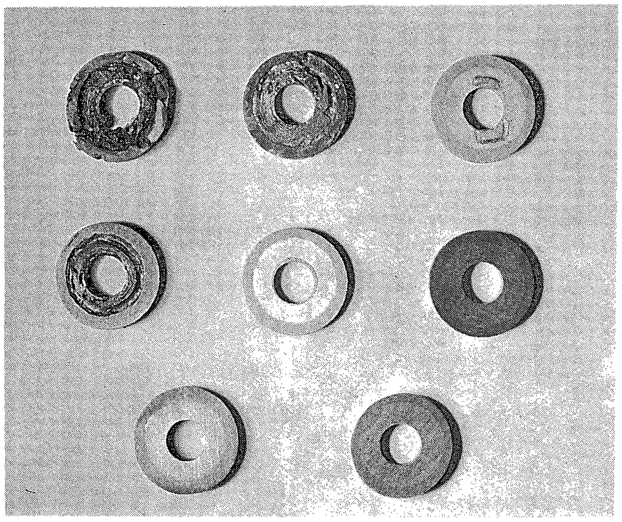
F475B E_c=298	F474B E_c=268	F473B E_c=235	F470B E_c=272
F475A E_c=152	F474A E_c=205	F473A E_c=165	F470A E_c=248

FIGURE 8 — Crevice corrosion after 6 day exposure to oxygenated 10% ferric chloride at 21 ± 1 C. Specimen size : ½ inch diameter.

immersion test in oxygenated, synthetic seawater at 121 C (250 F) are illustrated in Figures 11 and 12. All alloys studied exhibited crevice attack and/or pitting. Alloys with low values of E_c^{scr}, such as F389 (Figure 11) and 316 SS and F463B (Figure 12), showed no crevice attack but suffered severe pitting at the exposed edges of the specimen. E-Brite 26-1 (Figure 12) was severely attacked under the crevice.

Desalination Plant Tests

Five exposure tests have been performed at the OSW

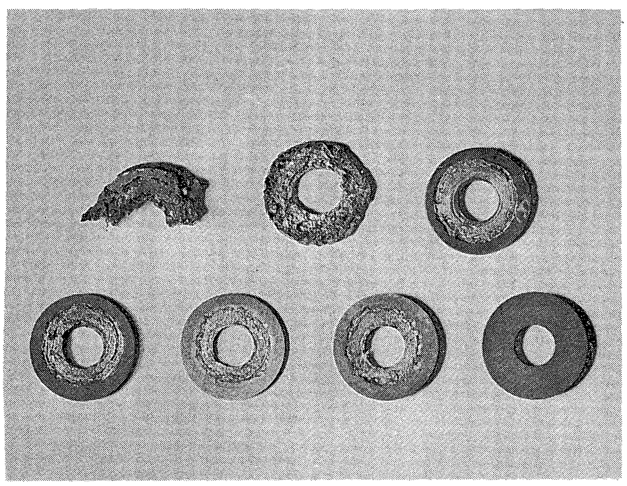


F295B Fe-20Cr-1Mo E_c=80 | F296B Fe-20Cr-3Mo E_c=138 | F297B Fe-20Cr-5Mo E_c=285

F291B Fe-25Cr-1Mo E_c=120 | F292B Fe-25Cr-3Mo E_c=262 | F293B Fe-25Cr-5Mo E_c=440

F223B Fe-30Cr-1Mo E_c=160 | F224B Fe-30Cr-3Mo E_c=500

FIGURE 9 — Crevice corrosion after 6 day exposure to oxygenated 10% ferric chloride at 21 ± 1 C. Specimen size: ½ inch diameter.

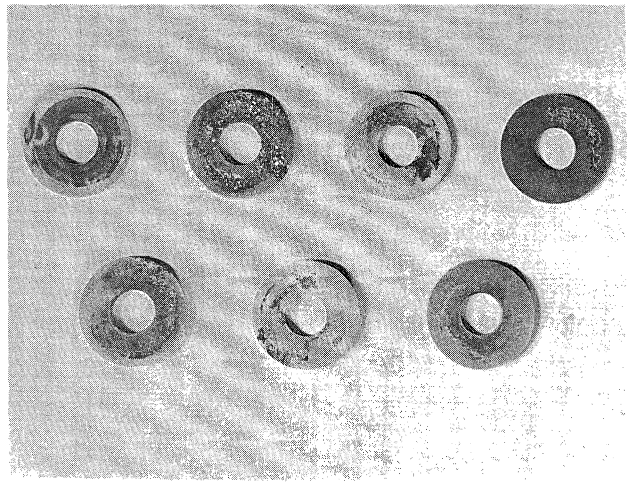


F312B Fe-17Cr-3Mo E_c=130 | F310B Fe-14Cr-5Mo E_c=172 | F313 Fe-17Cr-5Mo E_c=202

F314B Fe-25Cr E_c=-90 | F190B Fe-30Cr E_c=-50 | F191B Fe-35Cr E_c=-35 | F307B Fe-40Cr E_c=-10

FIGURE 10 — Crevice corrosion after 6 day exposure to oxygenated 10% ferric chloride at 21 ± 1 C. Specimen size: ½ inch diameter.

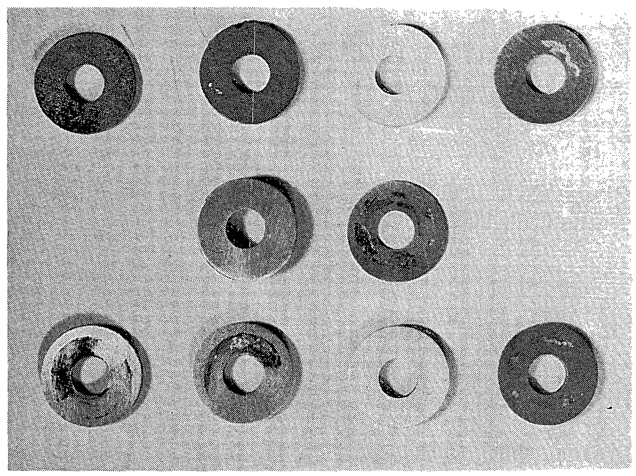
desalination plant test facility in Freeport, Texas using a single set of flat test coupons with dimensions 6 x 0.75 x 0.0625 inch. The corners of each specimen were held in Teflon supports to provide conditions for crevice attack, as illustrated in Figure 13. The conditions of each exposure and the measured corrosion rates are given in Table 6.



F389B F390B F294B F314B

F190B F191B F307B

FIGURE 11 — Crevice corrosion after 342 hour exposure to synthetic seawater (pH = 7.0) at 121 C and ~ 60 ppm oxygen.



F462B F463B F466B F470B

316SS E-Brite 26-1

F462A F463A F466A F470A

FIGURE 12 — Crevice corrosion after 342 hour exposure to synthetic seawater (pH = 7.0) at 121 C and ~ 60 ppm oxygen.

Following Run 1, the coupons were easily brushed clean. However, after Run 2, it was difficult to remove the deposit by simply brushing. The coupons were, therefore, chemically cleaned in a solution of inhibited HCl. Thereafter, all coupons from each test were chemically cleaned in inhibited HCl. From the mild steel results, it would appear that Runs 2, 3, and 4 did not really have the indicated amount of oxygen present. Run 5 apparently did have the indicated amount.

The results of Run 1, which are illustrated in Figure 14 and 15, show negligible attack on all test coupons with the exception of E-Brite 26-1. The weight loss data for Run 1 are plotted in Figure 4. The severity of the test environment was evidenced by the observation that mild steel

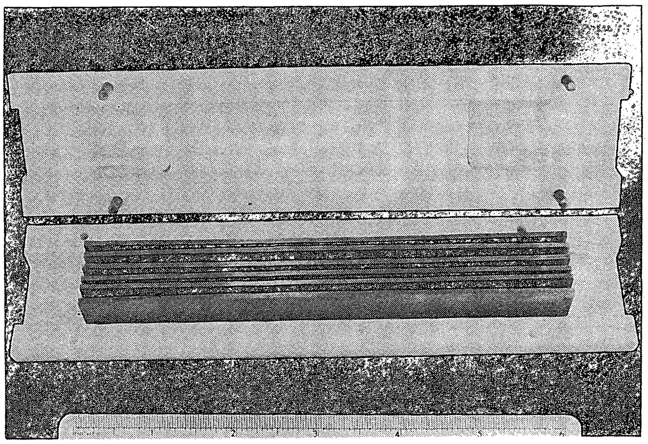


FIGURE 13 — Holder used for test coupons in desalination plant exposure tests.

TABLE 6 — Results of Desalination Plant Exposure Tests on Flat Coupons (Corrosion Rates Expressed in mpy)

Metal	Run 1	Run 2	Run 3	Run 4	Run 5
WE01	0.4	0.4	3	2	10
W501	0.02	0.008	0.02	0.5	3
W601	0.02	0.02	0.01	0.02	6
W701	0.01	0.007	0.004	0.03	2
W801	0.009	0.003	0.004	0.01	0.06
W581	0.02	0.06	0.07	0.07	2
W681	0.02	0.007	0.005	0.02	2
W781	0.008	0.0009	0.008	0	0.2
W881	0.007	0.005	0.01	0.01	0.04
WE61	0.3	0.3	2	1	9
W548	0.03	0.01	0.005	0.01	0.4
W648	0.02	0.009	0.006	0.01	0.2
W748	0.02	0.009	0.006	0	0.1
W848	0.02	0.005	0.006	0	0.004
1010 MS	35	12	7	7	63
SS316	—	0.1	0.1	0.2	2

Run 1 = Deaerated, 7 pH, 250 F, 30 days, ~ 4.0 fps velocity. Run 2(1) = ~ 75 ppb dissolved O_2, 7 pH, 250 F, 30 days, ~ 4.0 fps velocity. Run 3(1) = 75 ppb dissolved O_2, 7 pH, 250 F, 30 days, ~ 4.0 fps velocity. Run 4(2) = 200 ppb dissolved O_2, 7 ph, 250 F, 9 days, ~ 4.0 fps velocity. Run 5 = 200 ppb dissolved O_2, 7 pH, 250 F, 42 days, ~ 4.0 fps velocity.

(1) Mild steel results show no, or little O_2.
(2) Run cut short due to equipment failure.

coupons, which were also present during the second test, gave corrosion rates of 35 mpy.

The same test coupons were subsequently re-exposed four more times as detailed in Table 6. The results of Run 5 are plotted in Figure 5, and the appearance of the coupons after the fifth exposure at 200 ppb dissolved oxygen illustrated in Figures 16 and 17.

A comparison of Figures 4 and 5 indicates that an increase in the level of dissolved oxygen from $\lesssim$15 ppb to ~ 200 ppb produces marked differences in the relative corrosion resistances of the five stainless steels. At the lower range of oxygen concentrations, only E-Brite 26-1 exhibits crevice corrosion, while W5, W6, W7, and W8 in every condition are all essentially resistant to crevice attack

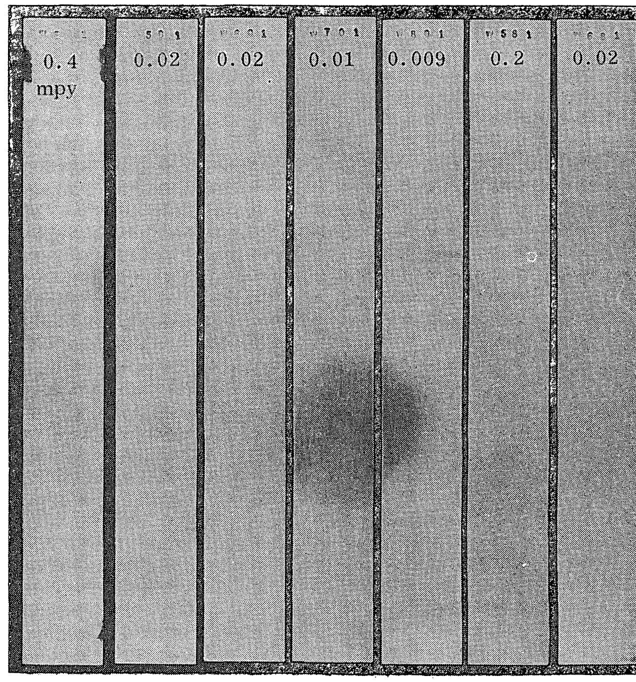


FIGURE 14 — Test coupons of alloys W5, W6, W7, W8, and E-Brite 26-1 (corrosion products removed) after 30 day exposure in Freeport desalination plant at 121 C, 4 ft/sec flow velocity, <5 ppb dissolved oxygen, <3 ppm CO_2, and pH = 7.0 ± 0.1. Initial surface finish pickled 5 minutes at 85 C in 10% HNO_3-0.5% HF. Note crevice attack on E-Brite 26-1 only.

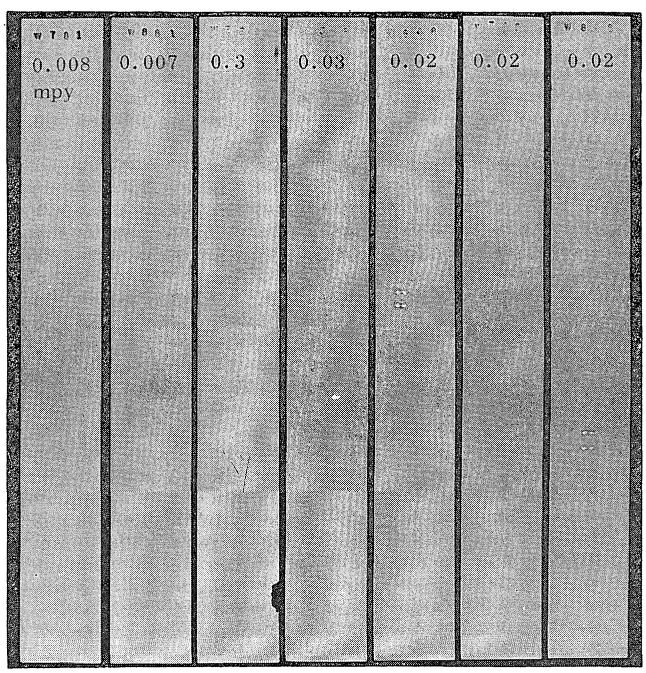


FIGURE 15 — Test coupons of alloys W5, W6, W7, W8, and E-Brite 26-1 (corrosion products removed) after 30 day exposure in Freeport desalination plant at 121 C, 4 ft/sec flow velocity, <5 ppb dissolved oxygen, <3 ppm CO_2, and pH = 7.0 ± 0.1. Initial surface finish pickled 5 minutes at 85 C in 10% HNO_3-0.5% HF. Note crevice attack on E-Brite 26-1 only.

and all corrode extremely slowly (~ 0.01 mpy). At the higher level of 200 ppb dissolved oxygen, all alloys except W8 are susceptible to crevice attack in the *as rolled*

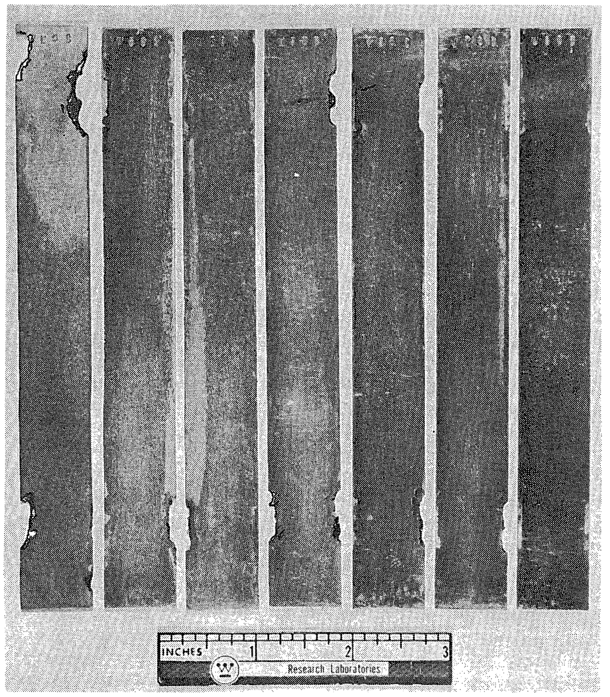


FIGURE 16 — Corrosion samples after Run 5 in desalination plant. Corrosion products have been removed.

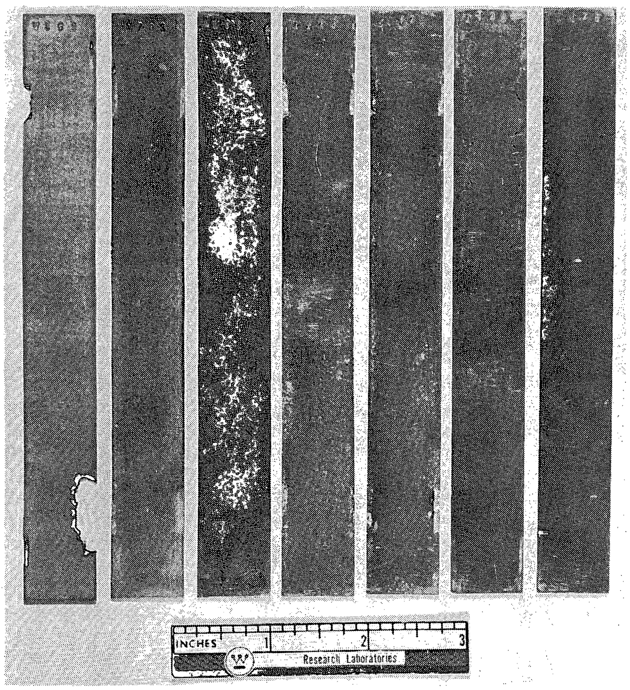


FIGURE 17 — Corrosion samples after Run 5 in desalination plant. Corrosion products have been removed.

condition. After a 427 C (800 F) x 1 hour heat treatment, only W7 and W8 are resistant to crevice attack. An anneal of 1093 C (2000 F) x 1 hour, however, produces resistance to crevice attack in W5, W6, W7, and W8, but the corrosion rates of the four alloys vary widely.

It is interesting to compare the results of the desalination plant exposures with the results of a 14 day autoclave test carried out on an identical series of specimens. The autoclave environment was synthetic seawater, at 121 C

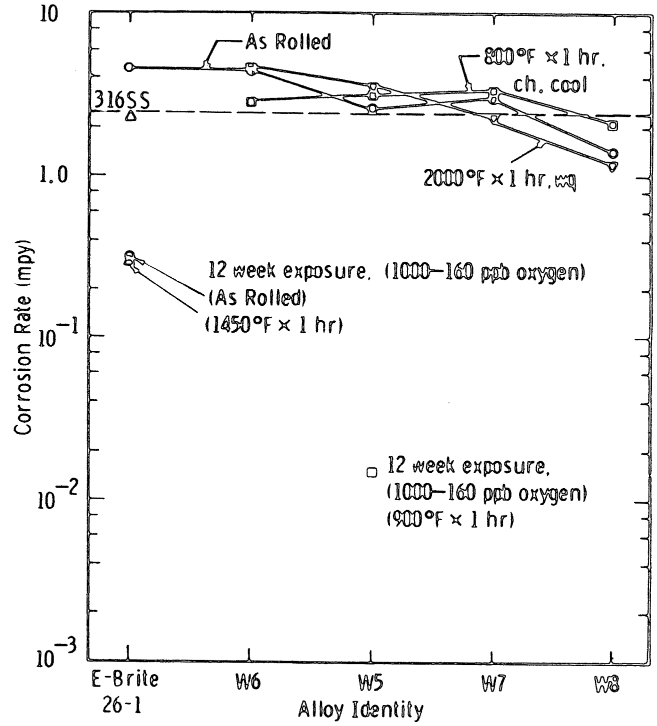


FIGURE 18 — Corrosion rates after 14 days autoclave exposure in synthetic seawater at 121 C, pH = 7.0, and dissolved oxygen level of 77 ppm.

with an initial dissolved oxygen content of 77 ppm. The results presented in Figure 18, indicate that at this high oxygen concentration all the stainless steels behave similarly and none of the alloys are resistant to crevice attack. Also included in Figure 18, are the results of a 12 week autoclave exposure of E-Brite 26-1 and alloy W5. The initial dissolved oxygen content was 1000 ppb which fell to 432 ppb after 2 weeks, was increased to 1000 ppb and fell to 160 ppb after 4 more weeks, was again increased to 1000 ppb and fell to 200 ppb after an additional 6 weeks. In this 12-week test with a lower average oxygen level, the alloy W5 was again resistant to crevice attack.

Metallographic Analyses

Duplicate specimens, prepared simultaneously with those specimens corrosion tested at the OSW Desalination Test Facility (Figures 16 and 17), were examined metallographically to correlate the results of the in-plant corrosion tests with specimen microstructure. The experimental alloys were evaluated in the following metallurgical conditions: as cold rolled, as cold rolled + 427 C/1 hour, and as cold rolled + 1093 C/1 hour. The standard reference material, E-Brite 26-1, was evaluated in the cold rolled and cold rolled + 788 C (1450 F)/1 hour conditions.

The microstructure of each specimen was examined in the rolling plane and in the plane through the sheet thickness (transverse to the rolling direction). Specimens were electrolytically etched in an aqueous solution containing 10% oxalic acid.

Typical microstructures for specimens W801, W881, and W848 are illustrated in Figure 19. The microstructures illustrated in Figure 19 are typical for each of the experimental alloys in each of the three metallurgical conditions. In the as cold-rolled and as cold-rolled + 427 C/1 hour

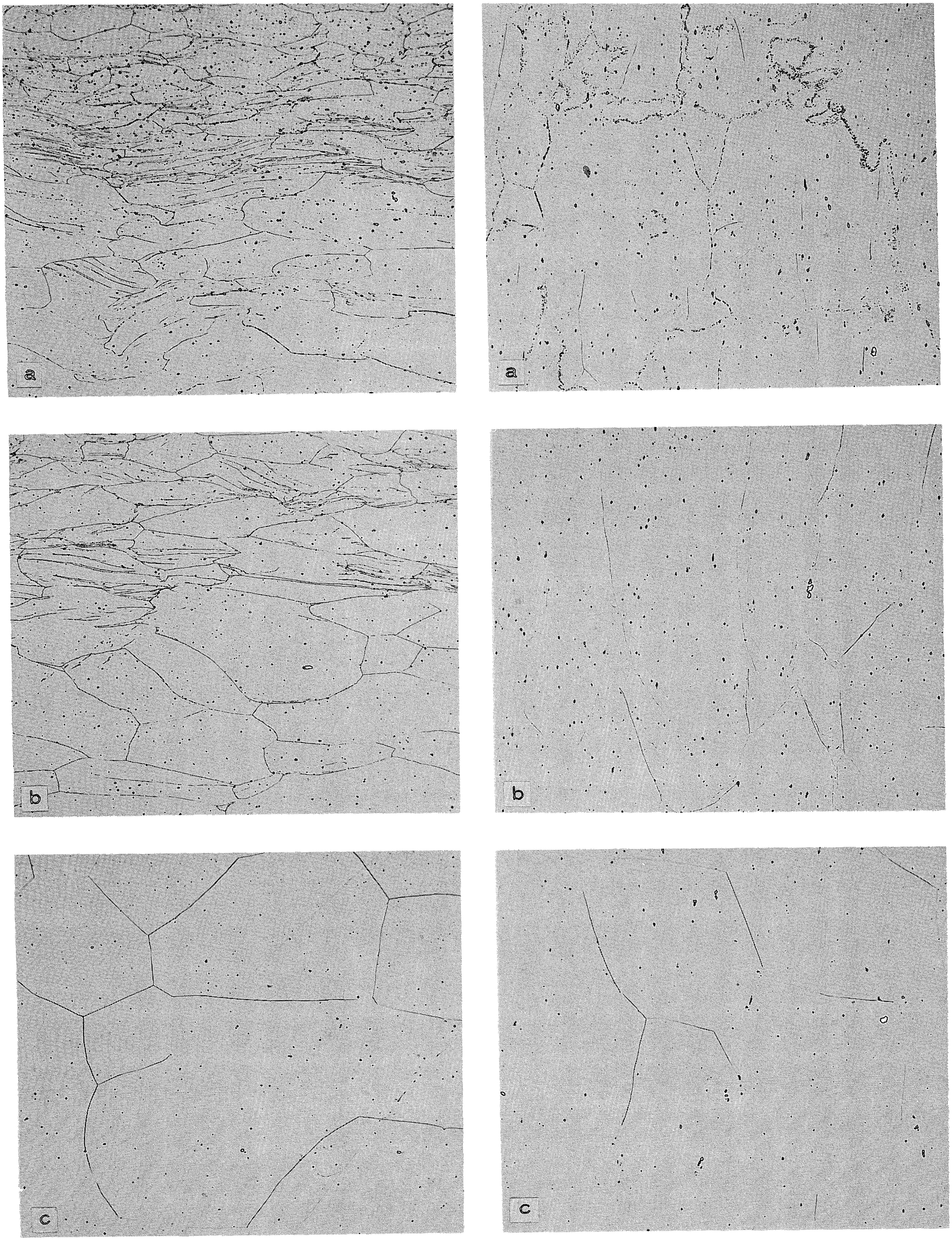


FIGURE 19 – Microstructures of experimental ferritic stainless steel alloys. Left column: transverse to rolling direction; right column: parallel to rolling plane. Electrolytic etch: 10% oxalic acid. (a) Alloy W802, as cold rolled, (b) alloy W882, as cold rolled + 427 C/1 hour, and (c) alloy W849, as cold rolled + 1093 C/1 hour.41X

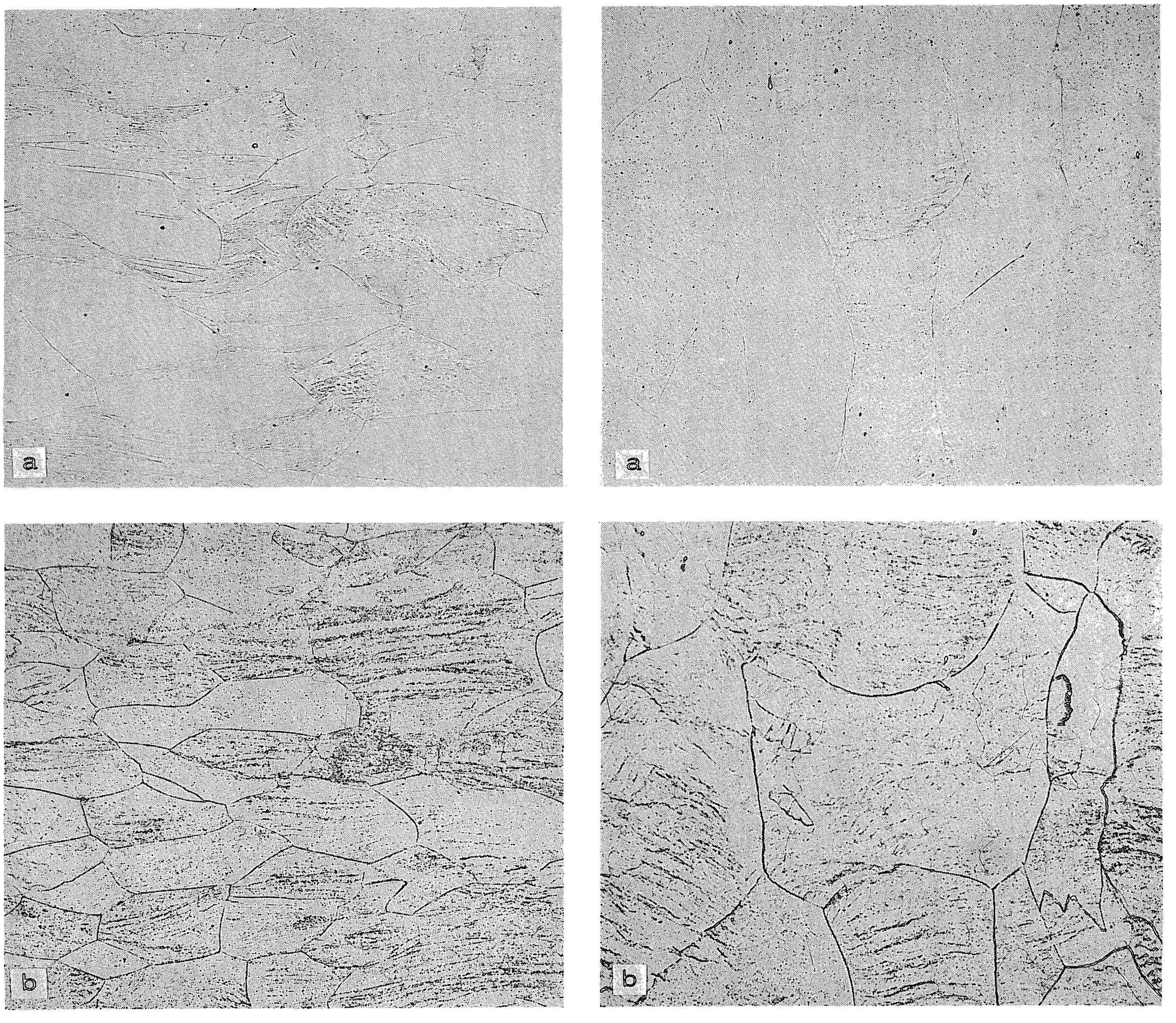


FIGURE 20 — Microstructure of E-Brite 26-1. Left column: transverse to rolling direction; right column: parallel to rolling plane. Electrolytic etch: 10% oxalic acid. (a) Alloy WE02, as cold rolled, and (b) alloy WE62, as cold rolled + 788 C/1 hour. 41X

conditions, grains are elongated in the rolling direction with the majority of deformation concentrated near the specimen surface. Transverse planes show multiple deformation bands within numerous grains oriented parallel to the rolling plane. The as cold-rolled microstructures (Figure 19a) show evidence of sigma-phase or chi-phase precipitate stringers located along prior hot rolled grain boundaries. However, the 427 C/1 hour heat treatment appears to dissolve these precipitates, as may be evidenced by Figure 19b. The cold rolled + 1093 C/1 hour microstructures show complete recrystallization has occurred within each of the experimental alloys. Grain boundaries and grain matrices are free of any form of precipitate.

Comparable microstructures for the reference specimens WE01 and WE61 are illustrated in Figure 20. The as cold rolled structure of E-Brite is similar to that observed in the experimental alloys. However, the microstructure for specimen WE61 is characterized by a continuous array of sigma-phase precipitates, located along prior deformed grain boundaries and slip bands within deformed grain matrices. This precipitate structure is super-imposed upon a fully recrystallized grain structure, free of grain boundary precipitates. The precipitate structure presumably formed during transient furnace heating through the equilibrium sigma-phase field.

The results of the corrosion tests correlate with the observed microstructures and chemical compositions of the test alloys. Crevice attack is most prevalent for the cold rolled specimens, the degree of attack decreasing with increasing Cr + Mo content. The 427 C/1 hour heat treatment dissolves precipitates formed in the prior hot rolled structure of the experimental alloys and subsequently decreases the susceptibility to crevice attack. The 1093 C/1 hour heat treatment eliminates residual cold work

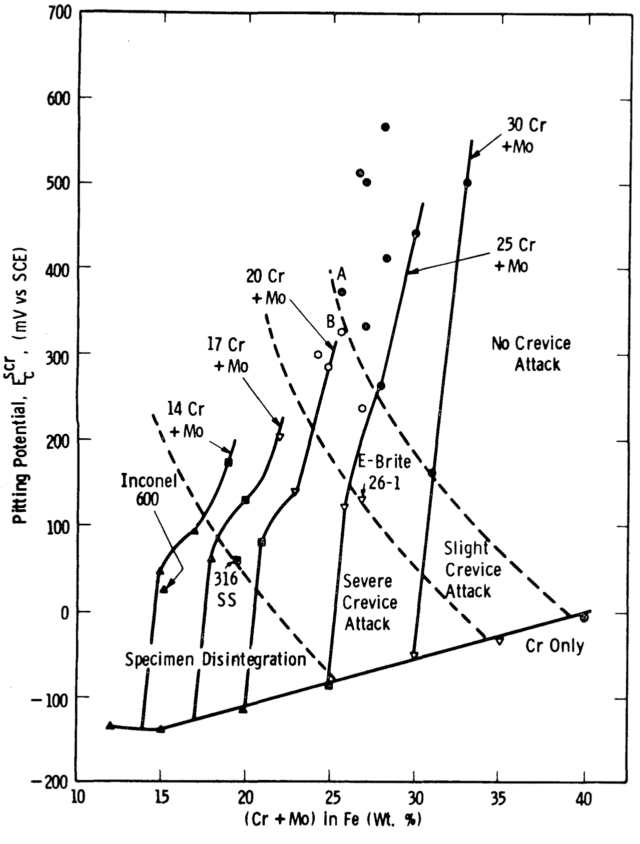


FIGURE 21 — Relationship between the critical pitting potentials (E_c^{scr} measured in deaerated synthetic seawater at 90 C, pH = 7.2 ± 0.2) of Fe-Cr-Mo based alloys and their resistance to crevice attack after a 6 day exposure in oxygenated, 10% ferric chloride at 21 ± 1 C.

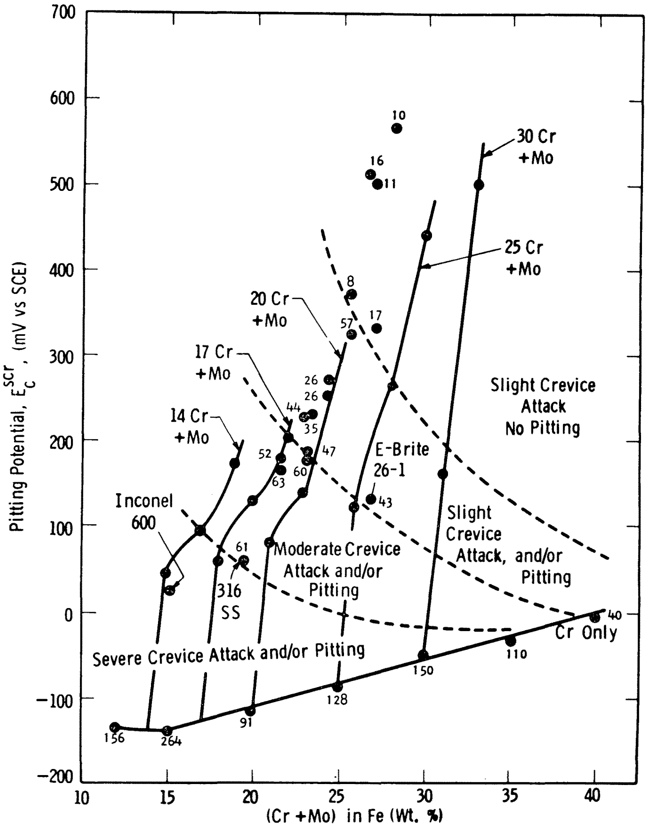


FIGURE 22 — Relationship between the critical pitting potentials (E_c^{scr} measured in deaerated synthetic seawater at 90 C, pH = 7.2 ± 0.2) of Fe-Cr-Mo based alloys and their resistance to crevice attack after a 14 day exposure to synthetic seawater at 121 C and ~ 60 ppm oxygen. The numbers associated with individual points indicate the weight losses expressed as gms x 10^{-4} after the 121 C exposure for samples with initial weights of ~ 1 gm.

through recrystallization and dissolves all precipitates, thus removing any possible solute concentration gradients in the vicinity of grain boundaries. This metallurgical condition yields the highest resistance to crevice corrosion as evidenced by corrosion test results and Figures 16 and 17.

Discussion

The relative pitting resistances of the alloys studied in this work have been based on critical pitting potential measurements (E_c^{scr}). Although similar techniques have been employed for many years as a means of rapidly screening alloys, the major efforts towards correlating pitting potential measurements with actual exposure in desalination plants have been rather limited. Consequently, a unique feature of the present work was that it allowed direct correlation between rapid laboratory tests and relatively long term in plant testing of a wide range of ferritic stainless steels.

In an attempt to correlate pitting potential (E_c^{scr}) with resistance to crevice attack, the results of the ferric chloride immersion tests have been superimposed on the E_c^{scr} data of Figure 2. The result is shown in Figure 21. It is found that the approximate contours of equal degrees of crevice attack resistance do not occur at constant values of E_c^{scr}. The contours instead suggest that the resistance to crevice attack (at least in the ferric chloride environment) is a function of both E_c^{scr} and the (Cr + Mo) concentration.

Point B in Figure 21, for example, indicates the E_c^{scr} value of an alloy (Fe-21.7Cr-4.0Mo) which shows some crevice attack in the ferric chloride test. If an (1.36Si + 1.44Ni) addition is made to this alloy, the E_c^{scr} is raised sufficiently, as shown by point A in Figure 21, to make the alloy immune to crevice attack in the same test. Conversely, as shown in Figure 8, if a ferritic alloy is heat treated such that the E_c^{scr} is lowered, then there is a concomitant decrease in the resistance to crevice attack.

The observations depicted in Figure 21 may be very significant in developing ferritic steels for desalination plant environments, or any other aqueous chloride environment. For example, the corresponding contours of equicrevice attack resistance for a particular environment could first be obtained by determining the E_c^{scr} value of those alloys containing Cr and Mo which are immune to crevice attack in the environment. With the crevice attack contour defined, a selection can be made of the composition which lies to the right of the contour and is optimized with respect to other desired properties.

An initial attempt has been made in this work to define such contours by means of the direct immersion tests in aerated seawater at 121 C. The results, presented in Figure 22, again indicate a dependence of crevice attack susceptibility on E_c^{scr} and (Cr + Mo) concentrations. A significant difference, however, is that the contours of equal crevice attack resistance in hot seawater are less dependent on the (Cr + Mo) concentration, but more dependent on the E_c^{scr} value than is observed for the corresponding results in ferric chloride tests.

Nevertheless, it should be noted that in order to obtain the maximum resistance to crevice attack, the data indicate that the chromium concentration should be as high as possible. Work aimed at substantiating this observation, by investigating the crevice attack resistance of high chromium alloys with relatively low pitting potential (E_c^{scr}) values, is now in progress.

All the samples tested for crevice attack resistance in synthetic seawater at 121 C exhibited crevice attack and/or pitting. This does not mean, however, that all the alloys have inadequate resistance to crevice attack for use in desalination plants, since the level of oxygen used in the laboratory tests (77 ppm) was more than three orders of magnitude greater than that normally encountered in actual service. The data does indicate, however, that in determining an adequate ferritic stainless steel for use in desalination plants, it will be absolutely essential to define the upper limit of oxygen concentration in the seawater. It is expected, therefore, that the application of relatively low cost ferritic steels will necessitate a very close control of oxygen level in the environment. The desalination plant tests described in this paper and others underway at Freeport, Texas were designed to evaluate the degree of oxygen control needed for application of any given alloy.

Summary and Conclusions

A study has been made of a series of high purity ferritic stainless steels based on the ternary alloy Fe-Cr-Mo with the objective of assessing the potential application of this class of material in condenser tubes of multistage flash evaporator desalination plants. Laboratory electrochemical tests and autoclave tests, which were used for preliminary alloy selection, have been directly compared to actual desalination plant exposure of identical alloys.

The following conclusions were reached:

1. Critical pitting potential measurements, obtained using a scratch method, E_c^{scr}, in synthetic seawater at 90 C, appear generally reliable in ranking the relative pitting resistance of ferritic stainless steels for use in desalination plants.

2. Autoclave tests using synthetic seawater at 121 C also give a ranking of crevice attack resistance which correlates well with actual desalination plant exposure.

3. Maximum resistance to crevice attack is obtained by raising both E_c^{scr} and the (Cr + Mo) concentration.

4. The presence of σ and/or χ phase in the microstructure of ferritic stainless steels markedly decreases their resistance to crevice corrosion in hot seawater. Correct heat treatment of the alloys before use is therefore essential.

5. High purity ferritic stainless steels can exhibit excellent resistance to both pitting and crevice attack in seawater at 121 C. However, a critical governing factor is the level of dissolved oxygen in the sea water. In deaerated seawater at 121 C many ferritic stainless steels have excellent corrosion resistance, but with increasing oxygen content the (Cr + Mo) concentration must be continually increased to maintain adequate corrosion resistance. Good resistance has been observed with ferritic stainless steels containing 21 to 23 wt% Cr and 3 to 5 wt% Mo, when exposed to 121 C seawater containing 200 to 500 ppb dissolved oxygen.

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